ALL ALUMINIUM ALLOY CONDUCTOR (AAAC)

AS PER IS -398-IV - 1994

Actual Area Sq. mm	tanding and Wire dia Nos./ mm	pprox. Overall dia mm	Approx . Kg / Km	Calculated Resistance Breaking Load Ohm/Km	Approx. Calculated at 20°C (Max) KN
15	3/2.50	5.39	40.15	2.3040	4.33
22	7/2.00	6.00	60.15	1.5410	6.45
34	7/2.50	7.50	94.00	0.9900	10.11
55	7/3.15	9.45	149.20	0.6210	16.03
80	7/3.81	11.43	218.26	0.4250	23.41
100	7/4.26	12.78	272.86	0.3390	26.26
125	19/2.89	14.45	342.51	0.2735	36.64
148	19/3.15	15.75	406.91	0.2290	43.50
173	19/3.40	17.00	474.02	0.1969	50.54
200	19/3.66	18.30	549.40	0.1710	58.66
232	19/3.94	19.70	636.67	0.1471	68.05
288	37/3.15	22.05	794.05	0.1182	84.71
346	37/3.45	24.15	952.56	0.0984	101.58
400	37/3.71	25.97	1101.63	0.0829	117.40
465	37/4.00	28.00	1280.50	0.0734	136.38
525	61/3.31	29.79	1448.39	0.0651	146.03
570	61/3.45	31.05	1573.71	0.0598	158.66
604	61/3.55	31.95	1666.00	0.0568	167.99
642	61/3.66	32.94	1771.36	0.0534	178.43
695	61/3.81	34.29	1919.13	0.0492	193.25
767	61/4.00	36.00	2115.54	0.0446	213.01

AS PER BS EN 50182 TYPE-AL3

Code Word	Sectional Area	Stranding		Diameter of Complete Conductor	Weight	Rated Strength	DC Resistance @ 20°C	Current Capacity	
		No. of Aluminium Wires	Individual wire diameter					@ 75°C	@ 85°C
	(mm^2)	(No.)	(mm)	(mm)	(Kg/Km)	KN	(Ω/Km)	(Ampere)	(Ampere)
BOX	18.8	7	1.85	5.55	51.4	5.55	1.748	84	99
ACACIA	23.8	7	2.08	6.24	64.9	7.02	1.3828	97	115
ALMOND	30.1	7	2.34	7.02	82.2	8.88	1.0926	111	133
CEDAR	35.5	7	2.54	7.62	96.8	10.46	0.9273	123	147
DEODAR	42.2	7	2.77	8.31	115.2	12.44	0.7797	136	163
FIR	47.8	7	2.95	8.85	130.6	14.11	0.6875	147	176
HAZEL	59.9	7	3.3	9.9	163.4	17.66	0.5494	167	202
PINE	71.6	7	3.61	10.8	195.6	21.14	0.4591	186	225
HOLLY	84.1	7	3.91	11.7	229.5	24.79	0.3913	205	248
WILLOW	89.7	7	4.04	12.1	245	26.47	0.3665	213	258
OAK	118.9	7	4.65	14	324.5	35.07	0.2767	251	306
MULBERRY	150.9	19	3.18	15.9	414.3	44.52	0.2192	288	354
ASH	180.7	19	3.48	17.4	496.1	53.31	0.183	320	394
ELM	211	19	3.76	18.8	579.2	62.24	0.1568	350	433
POPLAR	239.4	37	2.87	20.1	659.4	70.61	0.1387	376	467
SYCAMORE	303.2	37	3.23	22.6	835.2	89.4	0.1095	430	538
UPAS	362.1	37	3.53	24.7	997.5	106.82	0.0917	475	598
YEW	479	37	4.06	28.4	1319.6	141.31	0.0693	555	706
TOTARA	498.1	37	4.14	29	1372.1	146.93	0.0666	567	722
RUBUS	586.9	61	3.5	31.5	1622	173.13	0.0567	620	794
SORBUS	659.4	61	3.71	33.4	1822.5	194.53	0.0505	659	849
ARAUCARIA	821.1	61	4.14	37.3	2269.4	242.24	0.0406	740	962
REDWOOD	996.2	61	4.56	41	2753.2	293.88	0.0334	816	1072

AS PER AMERICAN SPECIFICATIONS ASTM B-399 - 2016

Nominal Alunium Sq. mm	tranding & Wire Dia No. / mm	Approx. Overall Diameter mm	Sectional Area Sq. mm	Approx Weight Kg/Km	pprox Calculated Breaking Load KN	Calculated Resistance at 20 °C (Max) Ohm/Km
16	7/1.70	5.10	15.89	43	4.44	2.0910
25	7/2.10	6.30	24.25	66	6.77	1.3720
35	7/2.50	7.50	34.36	94	9.6	0.9670
50	7/3.00	9.00	49.48	135	13.82	0.6716
50	19/1.80	9.00	48.35	133	13.5	0.6893
70	19/2.10	10.50	65.81	181	18.38	0.5079
95	19/2.50	12.50	93.27	256	26.05	0.3580
120	19/2.80	14.00	117.00	322	32.68	0.2853
150	37/2.25	15.00	147.11	406	41.09	0.2278
185	37/2.50	17.50	181.62	500	50.73	0.1842
240	61/2.25	20.30	242.54	670	67.74	0.1385
300	61/2.50	22.50	299.43	827	83.63	0.1119
400	61/2.89	26.00	400.14	1104	111.76	0.0841
500	61/3.23	29.10	499.83	1379	139.6	0.0671
625	91/2.96	32.60	626.20	1720	174.9	0.0538
800	91/3.35	36.90	802.09	2218	224.02	0.0420
1000	91/3.74	41.10	999.71	2767	279.22	0.0336